

# Coupled Pendulum

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## 1 Governing equations

Consider a coupled pendulum system with two identical pendulums connecting by a massless spring shown in Figure 1. Each pendulum consisting of a mass  $m$  suspended by a rigid massless rod of length  $L$ .

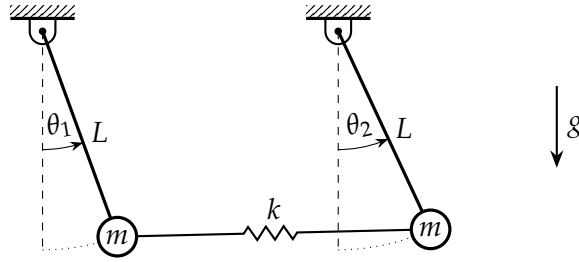


Figure 1: Pendulums coupled by a spring.

The kinetic and potential energies of the system are given by

$$T = \frac{1}{2}mL^2 (\dot{\theta}_1^2 + \dot{\theta}_2^2) \quad (1)$$

$$V = -mgL (\cos \theta_1 + \cos \theta_2) + \frac{1}{2}kx^2 \quad (2)$$

where  $x$  is the spring deformation. Thus, the Lagrangian  $\mathcal{L} = T - V$  is

$$\mathcal{L} = \frac{1}{2}mL^2 (\dot{\theta}_1^2 + \dot{\theta}_2^2) + mgL (\cos \theta_1 + \cos \theta_2) - \frac{1}{2}kx^2. \quad (3)$$

Now write

$$\frac{d}{dt} \left( \frac{\partial \mathcal{L}}{\partial \dot{\theta}_i} \right) - \frac{\partial \mathcal{L}}{\partial \theta_i} = 0, \quad i = 1, 2. \quad (4)$$

Hence, the equations of motion are

$$mL^2 \ddot{\theta}_i + mgL \sin \theta_i + kx \frac{\partial x}{\partial \theta_i} = 0. \quad (5)$$

Figure 2 illustrates the geometric model of the system. It demonstrates that the distance between two masses is

$$\bar{x} = \sqrt{(d - L(\sin \theta_1 - \sin \theta_2))^2 + L^2 (\cos \theta_1 - \cos \theta_2)^2} \quad (6)$$

where  $d$  is a distance between the pivots. Let  $\zeta$  be a natural length of the spring. Therefore, spring deformation is  $x = \bar{x} - \zeta$ . Hence,

$$\frac{\partial x}{\partial \theta_i} = \frac{\partial \bar{x}}{\partial \theta_i}. \quad (7)$$

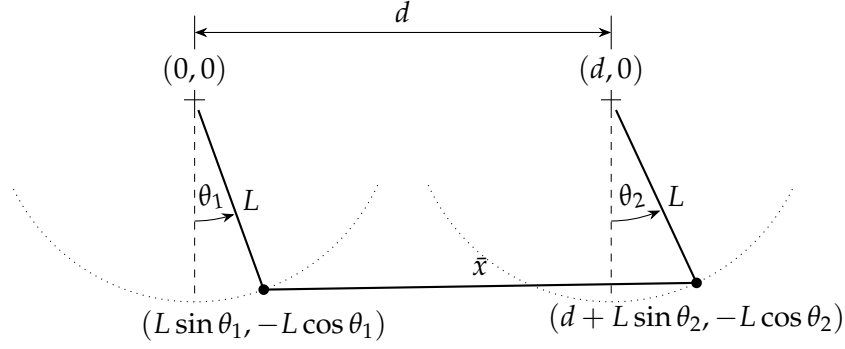


Figure 2: Geometry of the coupled pendulum system.

Therefore, we have

$$\frac{\partial \bar{x}}{\partial \theta_1} = \frac{-dL \cos \theta_1 + L^2 \sin (\theta_1 - \theta_2)}{\sqrt{(d - L(\sin \theta_1 - \sin \theta_2))^2 + L^2 (\cos \theta_1 - \cos \theta_2)^2}} \quad (8a)$$

$$\frac{\partial \bar{x}}{\partial \theta_2} = \frac{dL \cos \theta_2 - L^2 \sin (\theta_1 - \theta_2)}{\sqrt{(d - L(\sin \theta_1 - \sin \theta_2))^2 + L^2 (\cos \theta_1 - \cos \theta_2)^2}}. \quad (8b)$$

Substitute eq. eq8 into eq. (5), we obtain the equations of motion as

$$mL^2 \ddot{\theta}_1 + mgL \sin \theta_1 - k(\bar{x} - \xi) \left( \frac{dL \cos \theta_1 - L^2 \sin (\theta_1 - \theta_2)}{\bar{x}} \right) = 0 \quad (9a)$$

$$mL^2 \ddot{\theta}_2 + mgL \sin \theta_2 + k(\bar{x} - \xi) \left( \frac{dL \cos \theta_2 - L^2 \sin (\theta_1 - \theta_2)}{\bar{x}} \right) = 0. \quad (9b)$$

## 1.1 Linearization

If the amplitudes of oscillation are small, then  $\sin \theta_i \approx \theta_i$  and  $\cos \theta_i \approx 1$ . The equations of motion given by eq. (9) becomes

$$\mathbf{M} \ddot{\boldsymbol{\theta}} + \mathbf{K} \boldsymbol{\theta} = \mathbf{0} \quad (10)$$

where

$$\mathbf{M} = \begin{bmatrix} mL^2 & 0 \\ 0 & mL^2 \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} mgL + kL^2 & -kL^2 \\ -kL^2 & mgL + kL^2 \end{bmatrix}. \quad (11)$$

## 2 Normal modes

Consider the linear differential equations in eq. (10). Following the standard eigenvalue analysis [1, 2], the characteristic equation is

$$\det (\mathbf{K} - \omega^2 \mathbf{M}) = 0 \quad (12)$$

which yields

$$(mL^2)^2 \omega^4 - 2mL^2 (mgL + kL^2) \omega^2 + (mgL + kL^2)^2 - (kL^2)^2 = 0. \quad (13)$$

The solutions of eq. (13) give the natural frequencies

$$\omega_1 = \sqrt{\frac{g}{L}}, \quad \omega_2 = \sqrt{\frac{g}{L} + \frac{2k}{m}}. \quad (14)$$

The corresponding mode shapes  $\mathbf{u}_j = [u_{j1} \ u_{j2}]^T$  are obtained by substituting each natural frequency into the eigenvalue problem

$$(\mathbf{K} - \omega_j^2 \mathbf{M}) \mathbf{u}_j = \mathbf{0}, \quad j = 1, 2. \quad (15)$$

For the first mode, substituting  $\omega_1$  into eq. (15) gives

$$\begin{bmatrix} kL^2 & -kL^2 \\ -kL^2 & kL^2 \end{bmatrix} \begin{bmatrix} u_{11} \\ u_{12} \end{bmatrix} = \mathbf{0} \Rightarrow u_{11} = u_{12}. \quad (16)$$

Taking  $u_{11} = 1$ , the first mode shape is

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \quad (17)$$

This represents in-phase motion. Both pendulums move together and the spring remains unstretched, as shown in Figure 3(a).

For the second mode, substituting  $\omega_2$  into eq. (15) gives

$$\begin{bmatrix} -kL^2 & -kL^2 \\ -kL^2 & -kL^2 \end{bmatrix} \begin{bmatrix} u_{21} \\ u_{22} \end{bmatrix} = \mathbf{0} \Rightarrow u_{21} = -u_{22}. \quad (18)$$

Taking  $u_{21} = 1$ , the second mode shape is

$$\mathbf{u}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \quad (19)$$

This represents the out-of-phase motion. The pendulums oscillate in opposite directions, causing the spring to stretch and compress periodically, as shown in Figure 3(b)

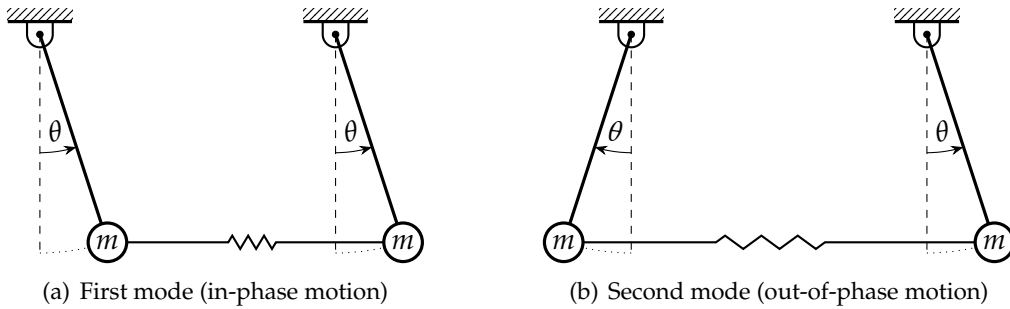


Figure 3: Normal modes of the coupled pendulum system.

The mode shapes are then normalized with respect to the mass matrix such that

$$\boldsymbol{\phi}_j^T \mathbf{M} \boldsymbol{\phi}_j = 1 \quad (20)$$

where  $\boldsymbol{\phi}_j = c_j \mathbf{u}_j$  denotes the mass-normalized mode shape and the constant  $c_j$  is given by

$$c_j = \frac{1}{\sqrt{\mathbf{u}_j^T \mathbf{M} \mathbf{u}_j}}. \quad (21)$$

Applying this to the first and second modes yields

$$c_1 = c_2 = \frac{1}{\sqrt{2mL^2}} \quad (22)$$

and the mass-normalized mode shapes are

$$\boldsymbol{\phi}_1 = \frac{1}{\sqrt{2mL^2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \boldsymbol{\phi}_2 = \frac{1}{\sqrt{2mL^2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \quad (23)$$

Therefore, the modal matrix  $\boldsymbol{\Phi} = [\boldsymbol{\phi}_1 \ \boldsymbol{\phi}_2]$  is

$$\boldsymbol{\Phi} = \frac{1}{\sqrt{2mL^2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}. \quad (24)$$

### 3 System response

The modal superposition is employed to determine the system response. The modal matrix  $\boldsymbol{\Phi}$  satisfies the orthogonality conditions

$$\boldsymbol{\Phi}^T \mathbf{M} \boldsymbol{\Phi} = \mathbf{I} \quad (25a)$$

$$\boldsymbol{\Phi}^T \mathbf{K} \boldsymbol{\Phi} = \boldsymbol{\Lambda} \quad (25b)$$

where  $\mathbf{I}$  is an identity matrix and  $\boldsymbol{\Lambda} = \text{diag}(\omega_1^2, \omega_2^2)$  is a diagonal matrix of squared natural frequencies. Using the coordinate transformation

$$\boldsymbol{\theta}(t) = \boldsymbol{\Phi} \boldsymbol{\eta}(t) \quad (26)$$

where  $\boldsymbol{\eta} = [\eta_1 \ \eta_2]^T$  is the modal position vector, the equations of motion become

$$\mathbf{I} \ddot{\boldsymbol{\eta}} + \boldsymbol{\Lambda} \boldsymbol{\eta} = \mathbf{0} \quad (27)$$

or in component form,

$$\ddot{\eta}_j + \omega_j^2 \eta_j = 0. \quad (28)$$

The system parameters are taken as  $m = 1 \text{ kg}$ ,  $k = 10 \text{ N/m}$ , and  $L = 1 \text{ m}$ , throughout this section unless otherwise specified. These parameters yield the natural frequencies of  $\omega_1 = 3.1316 \text{ rad/s}$  and  $\omega_2 = 5.4595 \text{ rad/s}$ .

#### 3.1 Free vibration response

The solution to eq. (28) is

$$\eta_j(t) = \eta_j(0) \cos(\omega_j t) + \frac{\dot{\eta}_j(0)}{\omega_j} \sin(\omega_j t) \quad (29)$$

where the initial conditions in modal coordinates are determined by

$$\eta_j(0) = \boldsymbol{\phi}_j^T \mathbf{M} \boldsymbol{\theta}(0) \quad \text{or} \quad \boldsymbol{\eta}(0) = \boldsymbol{\Phi}^T \mathbf{M} \boldsymbol{\theta}(0) \quad (30)$$

$$\dot{\eta}_j(0) = \boldsymbol{\phi}_j^T \mathbf{M} \dot{\boldsymbol{\theta}}(0) \quad \text{or} \quad \dot{\boldsymbol{\eta}}(0) = \boldsymbol{\Phi}^T \mathbf{M} \dot{\boldsymbol{\theta}}(0). \quad (31)$$

The free vibration response in physical coordinates is then obtained via eq. (26). If the initial conditions are aligned with the  $j$ -th mode shape, the response remains proportional to that mode shape throughout the motion at  $\omega_j$ .

**Case 1:**  $\theta(0) = [\theta_0 \ \theta_0]^T$  and  $\dot{\theta}(0) = [0 \ 0]^T$

The initial conditions are proportional to the first mode shape. Using eqs. (30) and (31) gives

$$\eta(0) = \sqrt{2mL^2} \begin{bmatrix} \theta_0 \\ 0 \end{bmatrix}, \quad \dot{\eta}(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (32)$$

Since  $\eta_2(0) = \dot{\eta}_2(0) = 0$ , only the first mode is excited. Using the initial conditions of eq. (32), the response in physical coordinates is

$$\theta(t) = \theta_0 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cos \omega_1 t. \quad (33)$$

This motion corresponds to the in-phase mode (see Section 2), as shown in Figure 4.

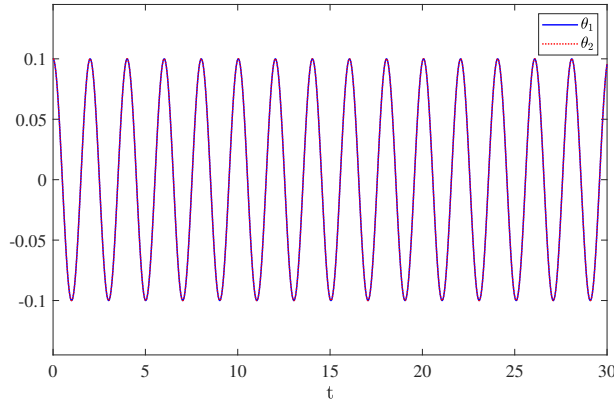


Figure 4: Free-vibration time response for case 1 when  $\theta_0 = 0.1$  rad/s.

**Case 2:**  $\theta(0) = [\theta_0 \ -\theta_0]^T$  and  $\dot{\theta}(0) = [0 \ 0]^T$

The initial conditions are proportional to the second mode shape. Using eqs. (30) and (31) gives

$$\eta(0) = \sqrt{2mL^2} \begin{bmatrix} 0 \\ \theta_0 \end{bmatrix}, \quad \dot{\eta}(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (34)$$

Since  $\eta_1(0) = \dot{\eta}_1(0) = 0$ , only the second mode is excited. Using the initial conditions of eq. (32), the response in physical coordinates is

$$\theta(t) = \theta_0 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \cos \omega_2 t. \quad (35)$$

This motion corresponds to the out-of-phase mode (see Section 2), as shown in Figure 5.

**Case 3:**  $\theta(0) = [\theta_0 \ 0]^T$  and  $\dot{\theta}(0) = [0 \ 0]^T$

The initial conditions are not proportional to either mode shape. Therefore, both modes are simultaneously excited. Using eqs. (30) and (31) gives

$$\eta(0) = \frac{\sqrt{2mL^2}}{2} \begin{bmatrix} \theta_0 \\ \theta_0 \end{bmatrix}, \quad \dot{\eta}(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (36)$$

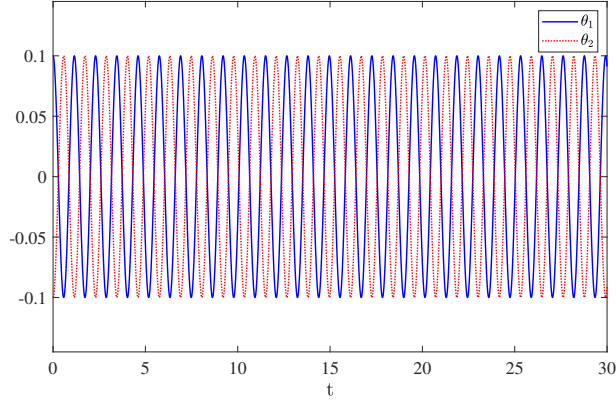


Figure 5: Free-vibration time response for case 2 when  $\theta_0 = 0.1$  rad/s.

Using the initial conditions of eq. (32), the response in physical coordinates is

$$\theta(t) = \frac{\theta_0}{2} \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cos \omega_1 t + \begin{bmatrix} 1 \\ -1 \end{bmatrix} \cos \omega_2 t \right\}. \quad (37)$$

The resulting motion is a superposition of both modes, as shown in Figure 6.

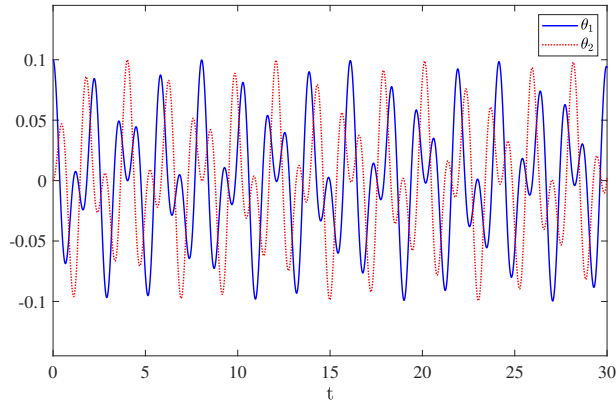


Figure 6: Free-vibration time response for case 3 when  $\theta_0 = 0.1$  rad/s.

### 3.1.1 Beat phenomenon

As observed in case 3, when the initial conditions are not aligned with a single mode shape, the response is a superposition of both modes. The energy is periodically transferred between two pendulums through a coupling spring.

Applying the product-to-sum trigonometric identities, eq. (37) can be rewritten as

$$\theta_1(t) = \theta_0 \cos \left( \frac{\omega_1 + \omega_2}{2} t \right) \cos \left( \frac{\omega_1 - \omega_2}{2} t \right) \quad (38)$$

$$\theta_2(t) = -\theta_0 \sin \left( \frac{\omega_1 + \omega_2}{2} t \right) \sin \left( \frac{\omega_1 - \omega_2}{2} t \right). \quad (39)$$

Thus, two distinct waves are obtained: a carrier and the beat. Each pendulum oscillates at the carrier frequency  $\omega_c$  with an amplitude that changes periodically at the beat frequency  $\omega_b$ . The carrier and beat frequencies are defined as

$$\omega_c = \frac{\omega_1 + \omega_2}{2} \quad \text{and} \quad \omega_b = \frac{|\omega_1 - \omega_2|}{2}. \quad (40)$$

This phenomenon is called *beating*. From eq. (14), the separation between  $\omega_1$  and  $\omega_2$  decreases as the coupling weakens ( $k \rightarrow 0^+$ ). This results in a longer beat period  $T_b = 2\pi/\omega_b$ . Due to a  $\pi/2$  phase difference between eqs. (38) and (39), one pendulum reaches its peak amplitude while the other is at rest. Figure 7 depicts the beating response for  $k = 2 \text{ N/m}$  under the same initial conditions as case 3.

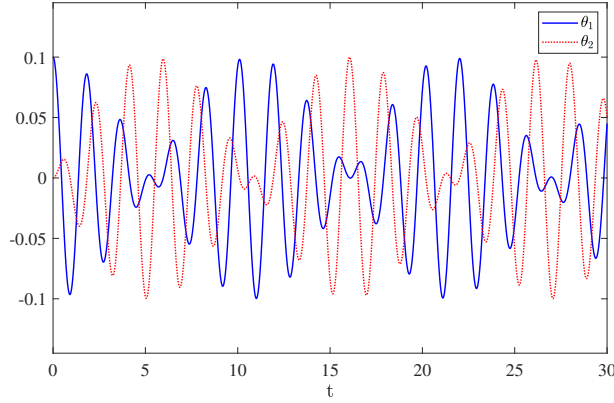


Figure 7: Free-vibration time response when  $k = 2 \text{ N/m}$  for initial conditions  $\theta(0) = [0.1 \ 0]^T$  and  $\dot{\theta}(0) = [0 \ 0]^T$ .

### 3.2 Forced response

Referring to Figure 1, consider external torques  $\tau_1$  and  $\tau_2$  applied to the pivots of the left and right pendulums, respectively. Under harmonic excitation, the equations of motion (10) become

$$\mathbf{M}\ddot{\theta} + \mathbf{K}\theta = \tau \sin \omega t \quad (41)$$

where  $\tau = [\tau_1 \ \tau_2]^T$ . The equations of motion in modal coordinates take the form

$$\mathbf{I}\ddot{\eta} + \mathbf{\Lambda}\eta = \gamma \sin \omega t \quad (42)$$

where  $\gamma = \Phi^T \tau$  is the modal torque amplitude vector. For zero initial conditions  $\eta(0) = \dot{\eta}(0) = \mathbf{0}$ , the full solution of each mode is

$$\eta_j(t) = \frac{\gamma_j}{\omega_j^2 - \omega^2} \left( \sin \omega t - \frac{\omega}{\omega_j} \sin \omega_j t \right), \quad \omega \neq \omega_j. \quad (43)$$

Transforming back to physical coordinates using eq. (26) gives the full response

$$\theta(t) = \sum_{j=1}^2 \phi_j \frac{\gamma_j}{\omega_j^2 - \omega^2} \left( \sin \omega t - \frac{\omega}{\omega_j} \sin \omega_j t \right). \quad (44)$$

For the torque input  $\tau = [\tau \ 0]^T$ , only the left pendulum is excited and the modal torque amplitudes are

$$\gamma_1 = \gamma_2 = \frac{\tau}{\sqrt{2mL^2}}. \quad (45)$$

The response in physical coordinates is

$$\theta(t) = \frac{\tau}{2mL^2} \left\{ \frac{1}{\omega_1^2 - \omega^2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \left( \sin \omega t - \frac{\omega}{\omega_1} \sin \omega_1 t \right) + \frac{1}{\omega_2^2 - \omega^2} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \left( \sin \omega t - \frac{\omega}{\omega_2} \sin \omega_2 t \right) \right\}. \quad (46)$$

Figure 8 shows the response for  $\tau = 1 \text{ N} \cdot \text{m}$  and  $\omega = 2 \text{ rad/s}$ .

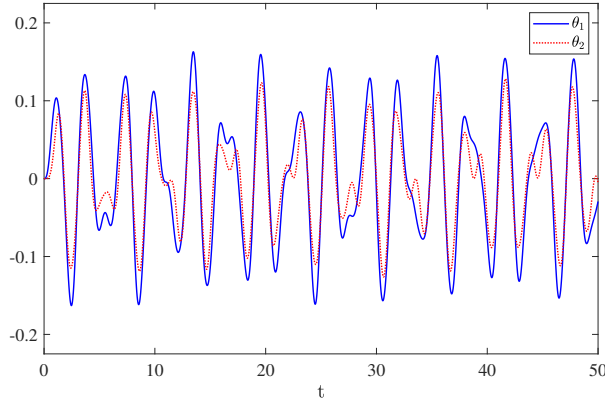


Figure 8: Time response to the torque input  $\tau(t) = [1 \ 0]^T \sin 2t$  with zero initial conditions.

### 3.2.1 Frequency response

The full solution of the forced response in eq. (43) consists of homogeneous and particular solutions. For the undamped system, the homogeneous response does not decay over time, while the particular solution oscillates at the excitation frequency  $\omega$ :

$$\eta_{j,p}(t) = \frac{\gamma_j}{\omega_j^2 - \omega^2} \sin \omega t. \quad (47)$$

Transforming back to physical coordinates gives

$$\theta_p(t) = \Theta \sin \omega t \quad (48)$$

where the amplitude vector is

$$\Theta = \sum_{j=1}^2 \phi_j \frac{\gamma_j}{\omega_j^2 - \omega^2}. \quad (49)$$

The amplitude  $\Theta_i$  becomes unbounded as  $\omega \rightarrow \omega_j$ .

For the torque input  $\tau = [\tau \ 0]^T$ , the amplitude ratios  $\Theta_i/\tau$  are

$$\frac{\Theta_1}{\tau} = \frac{1}{2mL^2} \left( \frac{1}{\omega_1^2 - \omega^2} + \frac{1}{\omega_2^2 - \omega^2} \right) \quad (50)$$

$$\frac{\Theta_2}{\tau} = \frac{1}{2mL^2} \left( \frac{1}{\omega_1^2 - \omega^2} - \frac{1}{\omega_2^2 - \omega^2} \right). \quad (51)$$

The sign of  $\Theta_i/\tau$  indicates whether the response is in phase or out of phase with the applied torque. Figure 9 shows the frequency responses  $\Theta_1/\tau$  and  $\Theta_2/\tau$ .

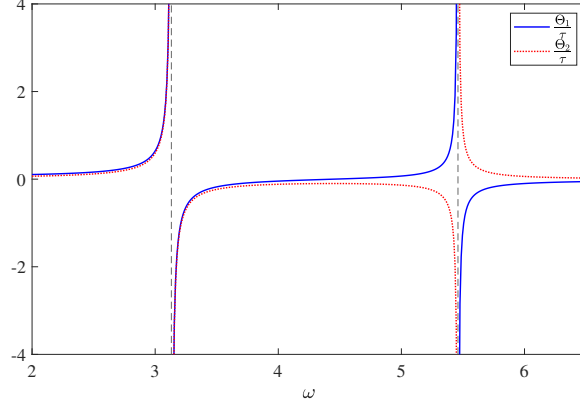


Figure 9: Frequency responses  $\Theta_1/\tau$  and  $\Theta_2/\tau$  for the torque input  $\tau = [\tau \ 0]^T$ .

### 3.3 Comparison with the nonlinear system

Validation of the linearized model is done by comparing its responses with the full nonlinear model. The nonlinear responses are obtained by solving the governing equations numerically with `ode45` in MATLAB, using the same system parameters as in the linear analysis. In addition,  $\xi$  and  $d$  are taken as 2 m.

The free vibration is first considered using the initial conditions of case 1 in Section 3.1. For a small initial angle, the linear and nonlinear responses are nearly identical over the first several cycles. For a larger initial angle, the nonlinear response gradually drifts out of phase with that of the linear model. This occurs because the oscillation period increases with amplitude [3, 4]. Figures 10(a) and 10(b) compare the responses of the two models for  $\theta_0 = 0.1$  rad and  $\theta_0 = 0.3$  rad, respectively.

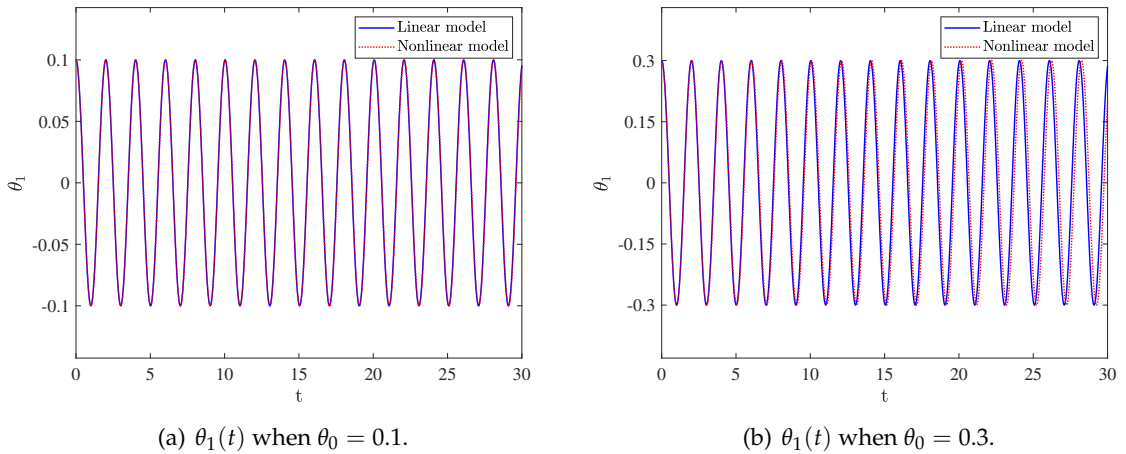


Figure 10: Comparison of the free-vibration responses of the linear and nonlinear models.

The response under harmonic excitation is next considered for  $\omega \approx \omega_1$ . Figure 11 compares the two responses under  $\tau = [0.5 \ 0]^T \sin 3t$  N · m. The responses of the linear and nonlinear

models are nearly identical during the first several cycles. As the amplitude increases, the nonlinear response grows larger and gradually departs from the linear prediction. This shows the range beyond which the linearized model no longer applies.

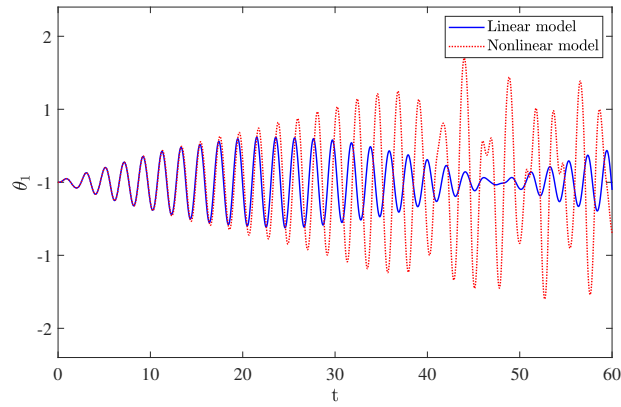


Figure 11: Comparison of the  $\theta_1(t)$  under  $\tau(t) = [0.5 \ 0]^T \sin 3t$  with zero initial conditions.

## 4 References

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